Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic probability**

Two fair six-sided dice are rolled. Find the probability their sum is 8.

Total outcomes: $n(S) = 36$. Favorable outcomes (sum $= 8$): $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$ — that's 5.

2 **[Foundational]** • **Target: Complement rule**

If $P(E) = 0.35$, find $P(E')$.

By Definition 6.2, $P(E') = 1 - P(E) = 1 - 0.35 = 0.65$.

3 **[Foundational]** • **Target: Permutations and combinations**

How many ways can a committee of 4 be chosen from 10 people?

Order doesn't matter, so use combinations: $\binom{10}{4} = \frac{10!}{4!\,6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$.

4 **[Foundational]** • **Target: Permutations and combinations**

How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?

Order matters, so use permutations: $P(6, 2) = \frac{6!}{4!} = 6 \cdot 5 = 30$.

5 **[Foundational]** • **Target: Permutations and combinations**

Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.

Theorem 6.1 says $\binom{6}{2} = \frac{P(6, 2)}{2!} = \frac{30}{2} = 15$ — combinations undercount permutations by exactly the $2!$ orderings of each chosen pair.

6 **[Foundational]** • **Target: Independent events and the addition rule**

Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

By Definition 6.4, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.4)(0.5) = 0.2$.

7 **[Foundational]** • **Target: Independent events and the addition rule**

A card is drawn from a standard 52-card deck. Find $P(\text{Queen or Spade})$.

$P(\text{Queen}) = \frac{4}{52}$, $P(\text{Spade}) = \frac{13}{52}$, and $P(\text{Queen and Spade}) = \frac{1}{52}$ (the Queen of Spades).

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Combinatorial Probability & Geometric Probability**

How many ways can a committee of 5 be chosen from 17 people? Show your complete algebraic work.

Final Answer: 6188. Order doesn't matter, so use combinations: $\binom{17}{5} = 6188$.

9 **[Intermediate]** • **Target: Basic probability**

How many ways can a committee of 2 be chosen from 18 people? Show your complete algebraic work.

Final Answer: 153. Order doesn't matter, so use combinations: $\binom{18}{2} = 153$.

10 **[Intermediate]** • **Target: Complement rule**

How many ways can a committee of 6 be chosen from 18 people? Show your complete algebraic work.

Final Answer: 18564. Order doesn't matter, so use combinations: $\binom{18}{6} = 18564$.

11 **[Intermediate]** • **Target: Permutations and combinations**

How many ways can a committee of 2 be chosen from 14 people? Show your complete algebraic work.

Final Answer: 91. Order doesn't matter, so use combinations: $\binom{14}{2} = 91$.

12 **[Intermediate]** • **Target: Independent events and the addition rule**

How many ways can a committee of 5 be chosen from 11 people? Show your complete algebraic work.

Final Answer: 462. Order doesn't matter, so use combinations: $\binom{11}{5} = 462$.

13 **[Intermediate]** • **Target: Discrete Random Variables & Variance**

How many ways can 2 distinct positions be filled from 12 candidates, where order matters? Show your complete algebraic work.

Final Answer: 132. Order matters, so use permutations: $P(12, 2) = \frac{12!}{10!} = 12 \times 11 = 132$.

14 **[Intermediate]** • **Target: Continuous Distributions & Probabilities**

How many ways can 4 distinct positions be filled from 9 candidates, where order matters? Show your complete algebraic work.

Final Answer: 3024. Order matters, so use permutations: $P(9, 4) = \frac{9!}{5!} = 9 \times 8 \times 7 \times 6 = 3024$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Conditional Probability & Joint Events • Cross-Lesson: 5.01 Introductory Calculus

How many ways can 4 distinct positions be filled from 14 candidates, where order matters? Show your complete algebraic work.

Final Answer: 24024. Order matters, so use permutations:
$P(14, 4) = \frac{14!}{10!} = 14 \times 13 \times 12 \times 11 = 24024$.

16 [Challenging] • Target: Combinatorial Probability & Geometric Probability • Cross-Lesson: 1.01 Complex Numbers

How many ways can 2 distinct positions be filled from 11 candidates, where order matters? Show your complete algebraic work.

Final Answer: 110. Order matters, so use permutations: $P(11, 2) = \frac{11!}{9!} = 11 \times 10 = 110$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Sequences & Series

How many ways can 4 distinct positions be filled from 15 candidates, where order matters? Show your complete algebraic work.

Final Answer: 32760. Order matters, so use permutations:
$P(15, 4) = \frac{15!}{11!} = 15 \times 14 \times 13 \times 12 = 32760$.

18 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

Events A and B are independent, with $P(A) = 0.1$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

Final Answer: $\frac{1}{25}$. By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.1)(0.4) = \frac{1}{25}$.

19 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

Events A and B are independent, with $P(A) = 0.2$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

Final Answer: $\frac{3}{50}$. By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.2)(0.3) = \frac{3}{50}$.

20 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.1$. Find $P(A \text{ and } B)$.

Final Answer: $\frac{1}{20}$. By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.5)(0.1) = \frac{1}{20}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic probability

A bag contains 9 yellow marbles, 5 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{1}{3}$

b. $\frac{3}{7}$ ✓

c. $\frac{10}{21}$

d. $\frac{5}{21}$

Total marbles: $n(S) = 21$. $P(\text{yellow}) = \frac{9}{21} = \frac{3}{7}$.

22 [Foundational] • Target: Complement rule

A bag contains 9 yellow marbles, 9 red marbles, and 6 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{5}{12}$

b. $\frac{11}{24}$

c. $\frac{1}{4}$

d. $\frac{3}{8}$ ✓

Total marbles: $n(S) = 24$. $P(\text{yellow}) = \frac{9}{24} = \frac{3}{8}$.

23 [Foundational] • Target: Permutations and combinations

A bag contains 7 blue marbles, 7 green marbles, and 3 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{9}{17}$

b. $\frac{8}{17}$

c. $\frac{3}{17}$

d. $\frac{7}{17}$ ✓

Total marbles: $n(S) = 17$. $P(\text{blue}) = \frac{7}{17}$.

24 [Foundational] • Target: Independent events and the addition rule

A bag contains 7 blue marbles, 7 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{19}$ ✓

b. $\frac{8}{19}$

c. $\frac{9}{19}$

d. $\frac{5}{19}$

Total marbles: $n(S) = 19$. $P(\text{blue}) = \frac{7}{19}$.

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25 [Foundational] • Target: Discrete Random Variables & Variance

A bag contains 4 blue marbles, 5 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{2}{7}$ ✓

b. $\frac{1}{2}$

c. $\frac{3}{7}$

d. $\frac{5}{14}$

Total marbles: $n(S) = 14$. $P(\text{blue}) = \frac{4}{14} = \frac{2}{7}$.

26 [Foundational] • Target: Continuous Distributions & Probabilities

A bag contains 9 blue marbles, 2 green marbles, and 4 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{4}{15}$

b. $\frac{2}{15}$

c. $\frac{2}{3}$

d. $\frac{3}{5}$ ✓

Total marbles: $n(S) = 15$. $P(\text{blue}) = \frac{9}{15} = \frac{3}{5}$.

27 [Foundational] • Target: Conditional Probability & Joint Events

A bag contains 3 red marbles, 3 blue marbles, and 4 green marbles. A marble is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{10}$ ✓

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$

Total marbles: $n(S) = 10$. $P(\text{red}) = \frac{3}{10}$.

28 [Foundational] • Target: Combinatorial Probability & Geometric Probability

How many ways can a committee of 6 be chosen from 7 people?

a. 1

b. 3

c. 11

d. 7 ✓

Order doesn't matter, so use combinations: $\binom{7}{6} = 7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Basic probability

How many ways can a committee of 5 be chosen from 16 people?

a. 4368 ✓ **b.** 8008

c. 4364 **d.** 4372

Order doesn't matter, so use combinations: $\binom{16}{5} = 4368$.

30 [Intermediate] • Target: Complement rule

How many ways can a committee of 3 be chosen from 14 people?

a. 364 ✓ **b.** 368

c. 1001 **d.** 360

Order doesn't matter, so use combinations: $\binom{14}{3} = 364$.

31 [Intermediate] • Target: Permutations and combinations

How many ways can a committee of 3 be chosen from 10 people?

a. 120 ✓ **b.** 116

c. 124 **d.** 210

Order doesn't matter, so use combinations: $\binom{10}{3} = 120$.

32 [Intermediate] • Target: Independent events and the addition rule

How many ways can a committee of 4 be chosen from 9 people?

a. 126 ✓ **b.** 122

c. 130 **d.** 132

Order doesn't matter, so use combinations: $\binom{9}{4} = 126$.

33 [Intermediate] • Target: Discrete Random Variables & Variance

How many ways can a committee of 5 be chosen from 17 people?

a. 6192 **b.** 12376

c. 6188 ✓ **d.** 6184

Order doesn't matter, so use combinations: $\binom{17}{5} = 6188$.

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34 [Intermediate] • Target: Continuous Distributions & Probabilities

How many ways can a committee of 2 be chosen from 14 people?

a. 87

b. 91 ✓

c. 364

d. 95

Order doesn't matter, so use combinations: $\binom{14}{2} = 91$.

35 [Intermediate] • Target: Conditional Probability & Joint Events

How many ways can 2 distinct positions be filled from 12 candidates, where order matters?

a. 66

b. 138

c. 132 ✓

d. 126

Order matters, so use permutations: $P(12, 2) = \frac{12!}{10!} = 12 \times 11 = 132$.

36 [Intermediate] • Target: Combinatorial Probability & Geometric Probability

How many ways can 4 distinct positions be filled from 14 candidates, where order matters?

a. 24018

b. 1001

c. 24030

d. 24024 ✓

Order matters, so use permutations: $P(14, 4) = \frac{14!}{10!} = 14 \times 13 \times 12 \times 11 = 24024$.

37 [Intermediate] • Target: Basic probability

How many ways can 4 distinct positions be filled from 15 candidates, where order matters?

a. 32760 ✓

b. 32766

c. 32754

d. 1365

Order matters, so use permutations: $P(15, 4) = \frac{15!}{11!} = 15 \times 14 \times 13 \times 12 = 32760$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

Events A and B are independent, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

a. $\frac{23}{100}$

b. $\frac{13}{100}$

c. $\frac{9}{50}$ ✓

d. $\frac{9}{10}$

By the independence rule, $P(A \text{ and } B) = P(A) \cdot P(B) = (0.6)(0.3) = \frac{9}{50}$.

39 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

Out of 27 equally likely boxes, 7 are labeled, 8 are sealed, and 7 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.

a. $\frac{2}{9}$

b. $\frac{10}{27}$

c. $\frac{8}{27}$ ✓

d. $\frac{5}{9}$

By the addition rule: $P(\text{labeled or sealed}) = \frac{7}{27} + \frac{8}{27} - \frac{7}{27} = \frac{8}{27}$.

40 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

Out of 37 equally likely numbers, 4 are even, 10 are prime, and 2 are both. One item is chosen at random. Find $P(\text{even or prime})$.

a. $\frac{14}{37}$

b. $\frac{10}{37}$

c. $\frac{12}{37}$ ✓

d. $\frac{15}{37}$

By the addition rule: $P(\text{even or prime}) = \frac{4}{37} + \frac{10}{37} - \frac{2}{37} = \frac{12}{37}$.

41 [Challenging] • Target: Discrete Random Variables & Variance • Cross-Lesson: 1.01 Complex Numbers

For the above variable X, what is the variance $\text{Var}(X)$?

a. 0.44 ✓

b. 0.50

c. 0.64

d. 0.84

$E(X^2) = 0^2(0.1) + 1^2(0.4) + 2^2(0.5) = 0.4 + 2.0 = 2.4$. $\text{Var}(X) = 2.4 - (1.4)^2 = 2.4 - 1.96 = 0.44$.

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42 [Challenging] • Target: Continuous Distributions & Probabilities • Cross-Lesson: 2.01 Sequences & Series

A continuous random variable X is uniformly distributed over the interval $[2, 10]$. What is $P(4 \le X \le 7)$?

a. $\frac{3}{8}$ ✓

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$

Probability density $f(x) = \frac{1}{10-2} = \frac{1}{8}$. $P(4 \le X \le 7) = \frac{7-4}{8} = \frac{3}{8}$.

43 [Challenging] • Target: Continuous Distributions & Probabilities • Cross-Lesson: 3.01 Functions & Graphs

In any continuous probability distribution, what is $P(X = c)$ for any single exact point c?

a. 0 ✓

b. 1

c. $\frac{1}{2}$

d. Undefined

For any continuous distribution, the probability of taking any single exact real value is 0 because the area of a single line is zero.

44 [Challenging] • Target: Conditional Probability & Joint Events • Cross-Lesson: 4.01 Matrices & Vectors

Given $P(A) = 0.6, P(B) = 0.3,$ and $P(A \cup B) = 0.7$, are events A and B independent?

a. **No, because** $P(A \cap B) = 0.20 \neq 0.18$ ✓

b. Yes, because $P(A \cap B) = 0.18$

c. Yes, because they are mutually exclusive

d. Cannot be determined

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.3 - 0.7 = 0.20$. For independence, $P(A)P(B) = 0.6 \times 0.3 = 0.18$. Since $0.20 \neq 0.18$, they are not independent.

45 [Challenging] • Target: Combinatorial Probability & Geometric Probability • Cross-Lesson: 5.01 Introductory Calculus

A point is chosen at random inside a square of side 10. What is the probability that it falls inside the inscribed circle?

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. $\frac{\pi}{8}$

d. $\frac{1}{2}$

Area of square $= 10^2 = 100$. Inscribed circle radius $= 5$, Area $= \pi(5^2) = 25\pi$. Probability $= \frac{25\pi}{100} = \frac{\pi}{4} \approx 0.785$.

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